

Municipal Law Skill

for Autonomous Agents

An agent-ready municipal law skill providing deterministic, citation-backed legal evidence.

NANDA Town · MIT Reality Hackathon 2026

<https://nanda-municipal-laws.vercel.app>

AI agents hallucinate municipal law



Internet folklore

"You may keep up to six hens."



NYC Health Code §161.19 (the real statute)

No such limit exists.

Only certain birds (roosters, ducks, geese, turkeys) are prohibited — with narrow exceptions. Nothing restricts hens at all.

Agents relying on an LLM's training data repeat exactly this kind of confident, wrong answer. Legal citations need to come from the actual code — not a model's memory.

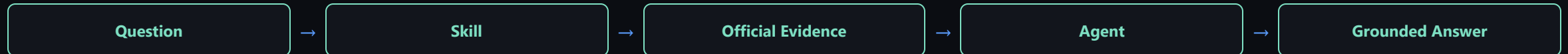
The capability

"A deterministic municipal law skill that any AI agent can invoke to obtain grounded, citation-backed legal evidence from the complete NYC Administrative and Health Codes."

Traditional RAG



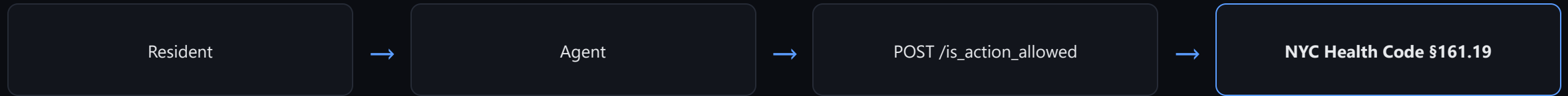
Municipal Law Skill



The service never generates answers. It only returns legal evidence.

Story 1 — A resident

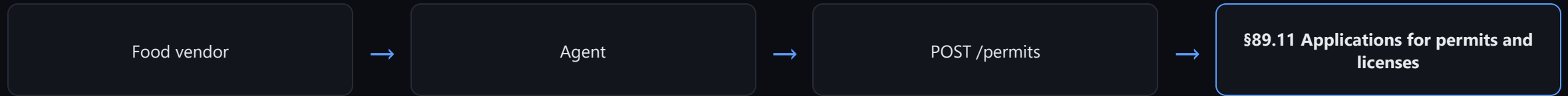
"Can I keep backyard chickens?"



"Yes, but roosters are prohibited citywide"

Story 2 — A food vendor

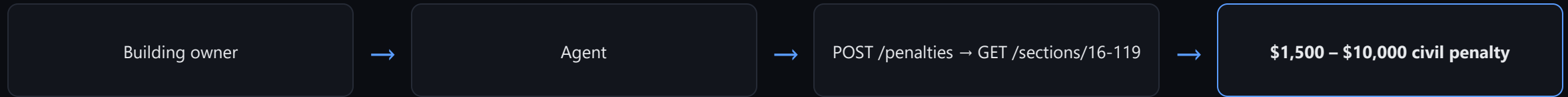
“What permit do I need to operate a mobile food cart?”



“You need a vending permit/license under §89.11 — fees are set by §17-308”

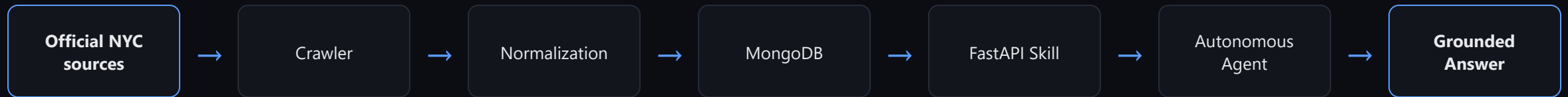
Story 3 — A building owner

“What's the penalty if I dump construction debris illegally?”



Real dollar figures, straight from the statute — not a vague “consult a lawyer.”

Architecture



The service never generates legal advice — only official text and citations. The last step is why it exists.

The complete corpus, not a sample

Entire NYC Administrative Code

32

Titles

4,781

Sections

Entire NYC Health Code

36

Articles

501

Sections

668 source documents · 10,702 searchable chunks · ≈55 MB total

Not a curated demo dataset.

Production issues discovered using live municipal data

- ▶ Naive substring search: "fine" matched inside "defined" — fixed with word-boundary matching everywhere
- ▶ 36 Health Code PDFs, each formatted slightly differently — parser normalized and verified section-by-section against real text
- ▶ The "6 hens" myth — a regression test locks in the true statute text, forever
- ▶ Hidden ingestion size limit silently truncated requests — found and fixed before it silently dropped data

Why not just ask an LLM?

LLM memory

- ✗ May be incomplete
- ✗ May be outdated
- ✗ Cannot prove provenance

Municipal Law Skill

- ✓ Official source
- ✓ Deterministic
- ✓ Citations
- ✓ Verifiable

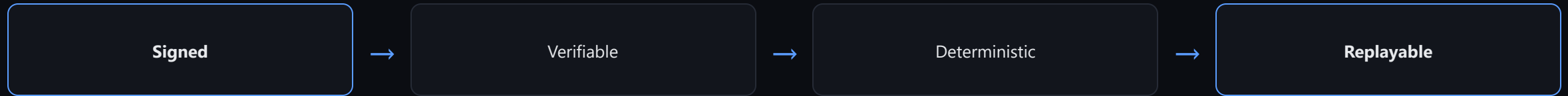
Agent-facing capabilities

Capability	Endpoint	Returns
Determine legality	/is_action_allowed	Allowed, conditions, citations, confidence
Search law	/search	Ranked legal evidence, filterable by source
Retrieve statute	/sections/{id}	Official, untruncated text
Find penalties	/penalties	Enforcement provisions
Find permits	/permits	Permit/license requirements

Also available, kept separate from the agent-facing surface above: /health, /version, /pubkey, /ingest — operational only, not part of the reasoning loop.

Trust & verification

Every response is cryptographically signed (Ed25519).



/pubkey lets any downstream agent verify offline that this service, not a relay or a cache, produced a citation. /version reports corpus freshness, so callers know exactly how stale the law is.

Why NANDA

“This transforms municipal law from static documents into a reusable capability that any NANDA agent can invoke.”

- ▶ Composable by design — the last hop in a chain started by another skill (a 311 complaint, a permit application, a code-enforcement lead)
- ▶ Any agent that can call an HTTP endpoint can invoke it, per SKILL.md

Any agent. Any language. Any framework. Just HTTP.

Agent-ready. Deterministic. Citation-backed.

- ✓ Complete NYC Administrative & Health Codes
- ✓ Deterministic, citation-backed legal evidence
- ✓ Live, agent-ready HTTP skill

<https://nanda-municipal-laws.vercel.app>

<https://github.com/swamx/nanda-municipal-laws>